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SKY130 VSDBabySoC Post-Route PVT Timing Analysis

OpenSTA Static Timing Analysis

1. Objective

To evaluate the post-route timing performance of the VSDBabySoC design across the available SKY130 HD process-voltage-temperature (PVT) corners using OpenSTA. The analysis aims to understand the effect of process variations, supply voltage, operating temperature, and routing parasitics on the timing performance of the design.

2. Timing Metrics

Worst Negative Slack (WNS) and Total Negative Slack (TNS) were obtained for each PVT corner using OpenSTA timing reports.

PVT Corner Description

PVT (Process-Voltage-Temperature) analysis evaluates the timing behavior of a digital circuit under different manufacturing process conditions, supply voltages, and operating temperatures. Analyzing multiple PVT corners helps determine whether the design meets its timing requirements under different operating conditions.

Timing Metrics

WNS (Worst Negative Slack): WNS represents the worst timing slack among the analyzed timing paths. A negative WNS indicates that at least one analyzed path violates its timing requirement. A WNS value of 0 ns indicates that no negative worst slack is reported.

TNS (Total Negative Slack): TNS represents the total negative slack across all violating timing paths. A TNS value of 0 ns indicates that no negative timing slack is reported in the analyzed paths.

3. Post-Route STA Setup

The VSDBabySoC design was analyzed after the routing stage using OpenSTA across the 16 SKY130 HD PVT timing corners. The post-route analysis considers the timing behavior of the routed design, including interconnect parasitics when extracted parasitic information is loaded into the timing analysis.

For each PVT corner, the corresponding SKY130 HD standard-cell Liberty timing library was selected. The PLL and DAC macro timing libraries, routed design netlist or database, SDC timing constraints, and extracted parasitic information were included as appropriate for the analysis flow.

Files and data used:

- SKY130 HD standard-cell Liberty timing libraries for the respective PVT corners
- avsdpll.lib – PLL macro timing library
- avsddac.lib – DAC macro timing library
- Post-route VSDBabySoC netlist or routed design database
- SDC timing constraint file

OpenSTA reports generated for each corner:

- report_checks
- report_wns
- report_tns

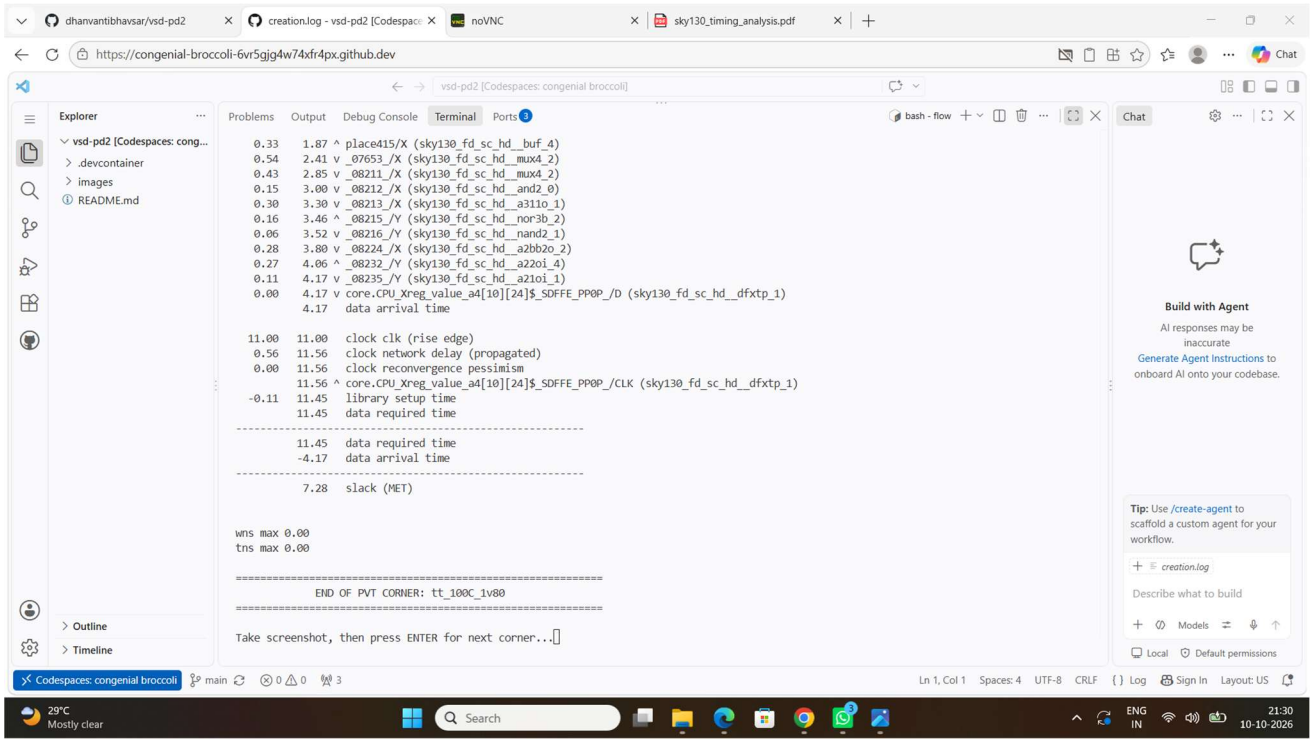
The reports were generated using the timing libraries, design data, constraints, and parasitic information appropriate to each PVT corner. The exact commands depend on the files produced by the OpenROAD post-route flow.

4. WNS and TNS Results Across PVT Corners

Corner	WNS (ns)	TNS (ns)
tt_025C_1v80	0.00	0.00
ff_100C_1v65	0.00	0.00
ff_100C_1v95	0.00	0.00
ff_n40C_1v56	0.00	0.00
ff_n40C_1v65	0.00	0.00
ff_n40C_1v76	0.00	0.00
ff_n40C_1v95	0.00	0.00
ss_100C_1v40	-0.34	-3.53
ss_100C_1v60	0	0
ss_n40C_1v28	-20.67	-5718.22
ss_n40C_1v35	-10.57	-1976.07
ss_n40C_1v40	-6.30	-736.46
ss_n40C_1v44	-3.77	-187.02
ss_n40C_1v60	0.00	0.00
ss_n40C_1v76	0.00	0.00
tt_100C_1v80	0.00	0.00

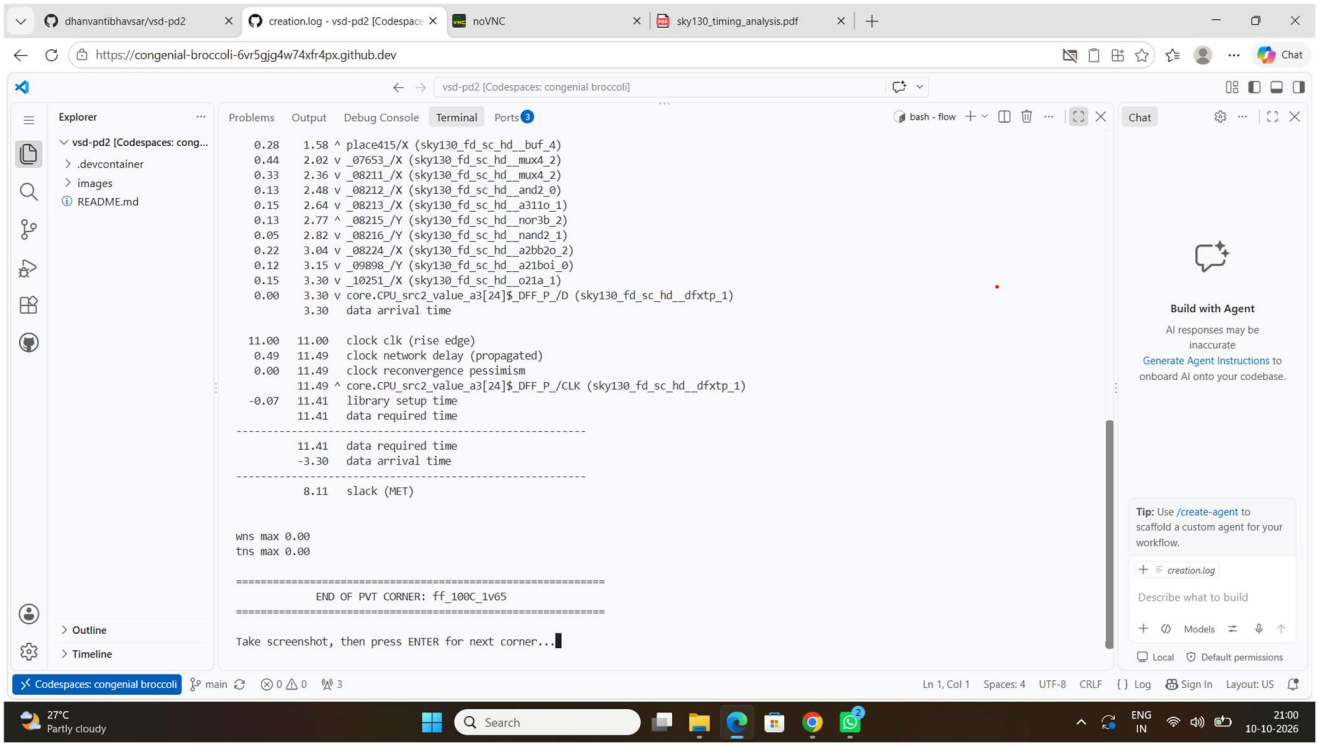
PVT Corner: tt_025C_1v80

This represents the typical-typical (TT) process corner at 25°C and 1.80 V supply voltage.



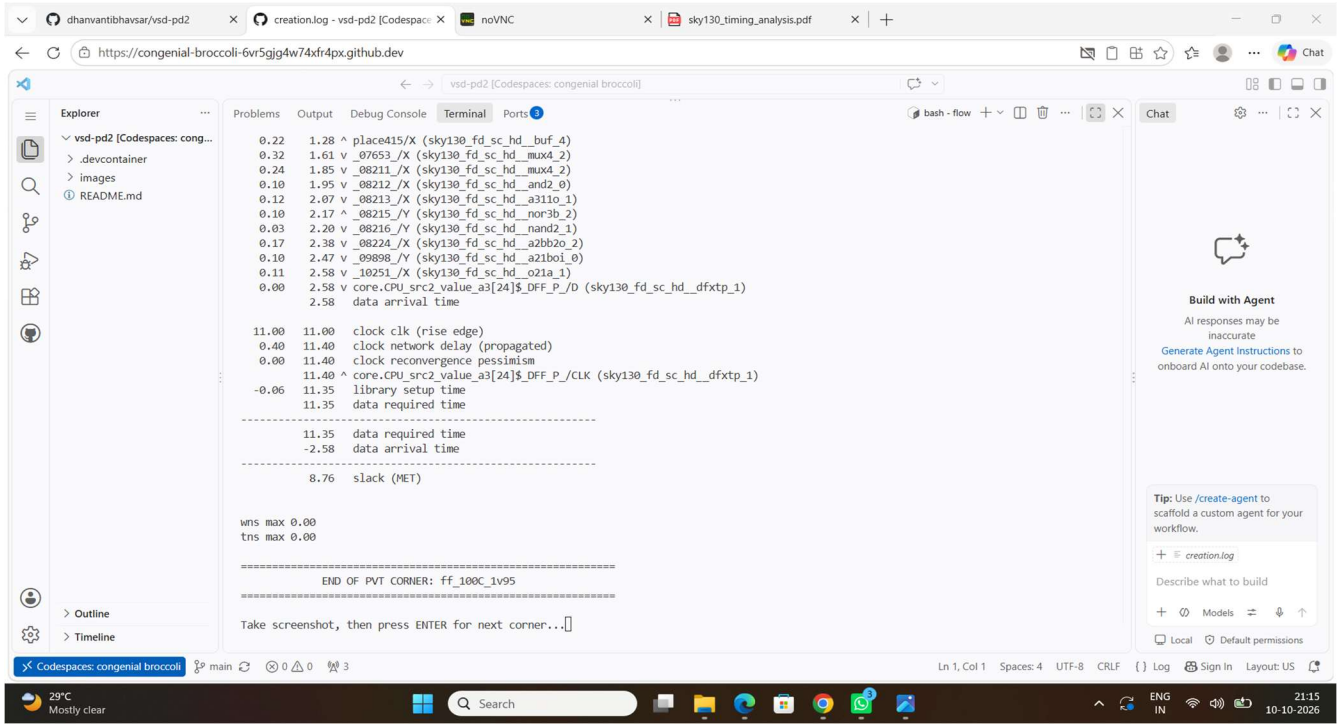
PVT Corner: ff_100C_1v65

This represents the fast-fast (FF) process corner at 100°C and 1.65 V supply voltage.



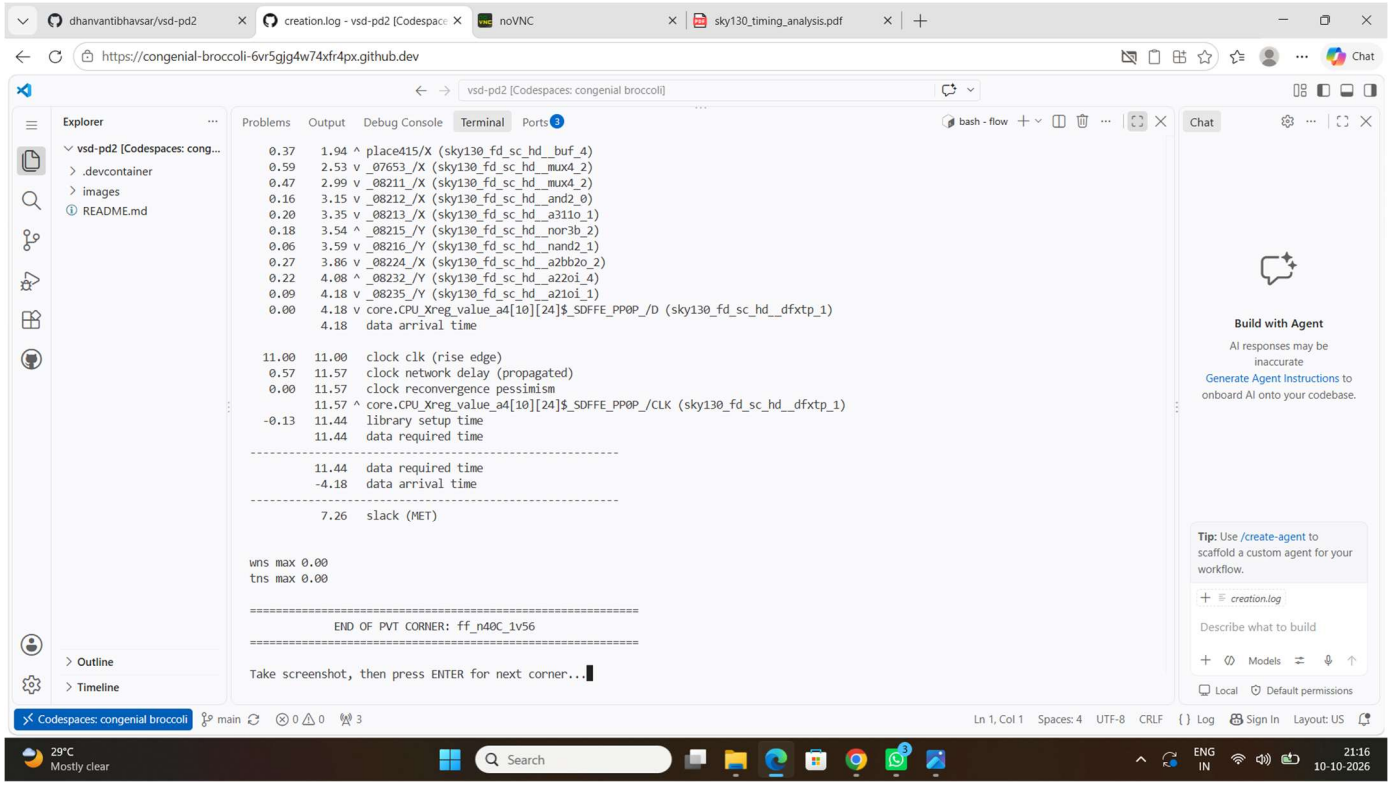
PVT Corner: ff_100C_1v95

This represents the fast-fast (FF) process corner at 100°C and 1.95 V supply voltage.



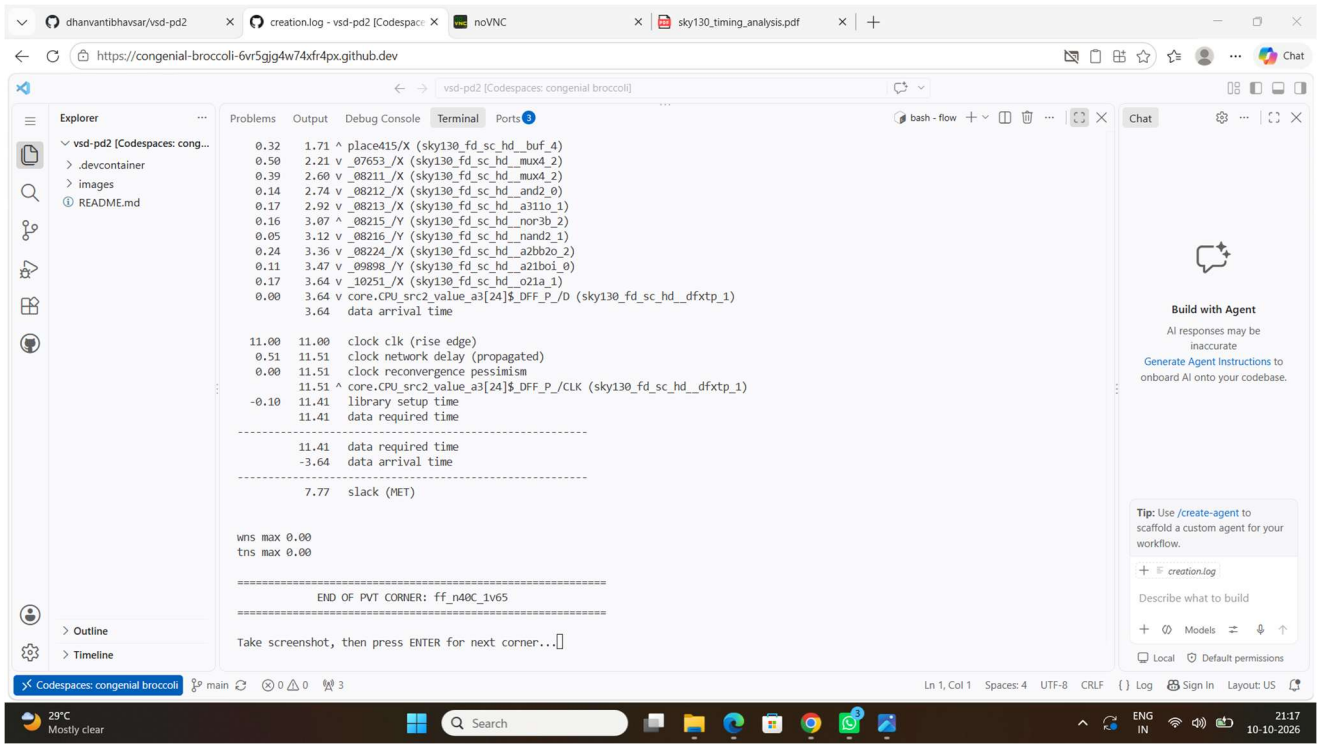
PVT Corner: ff_n40C_1v56

This represents the fast-fast (FF) process corner at −40°C and 1.56 V supply voltage.



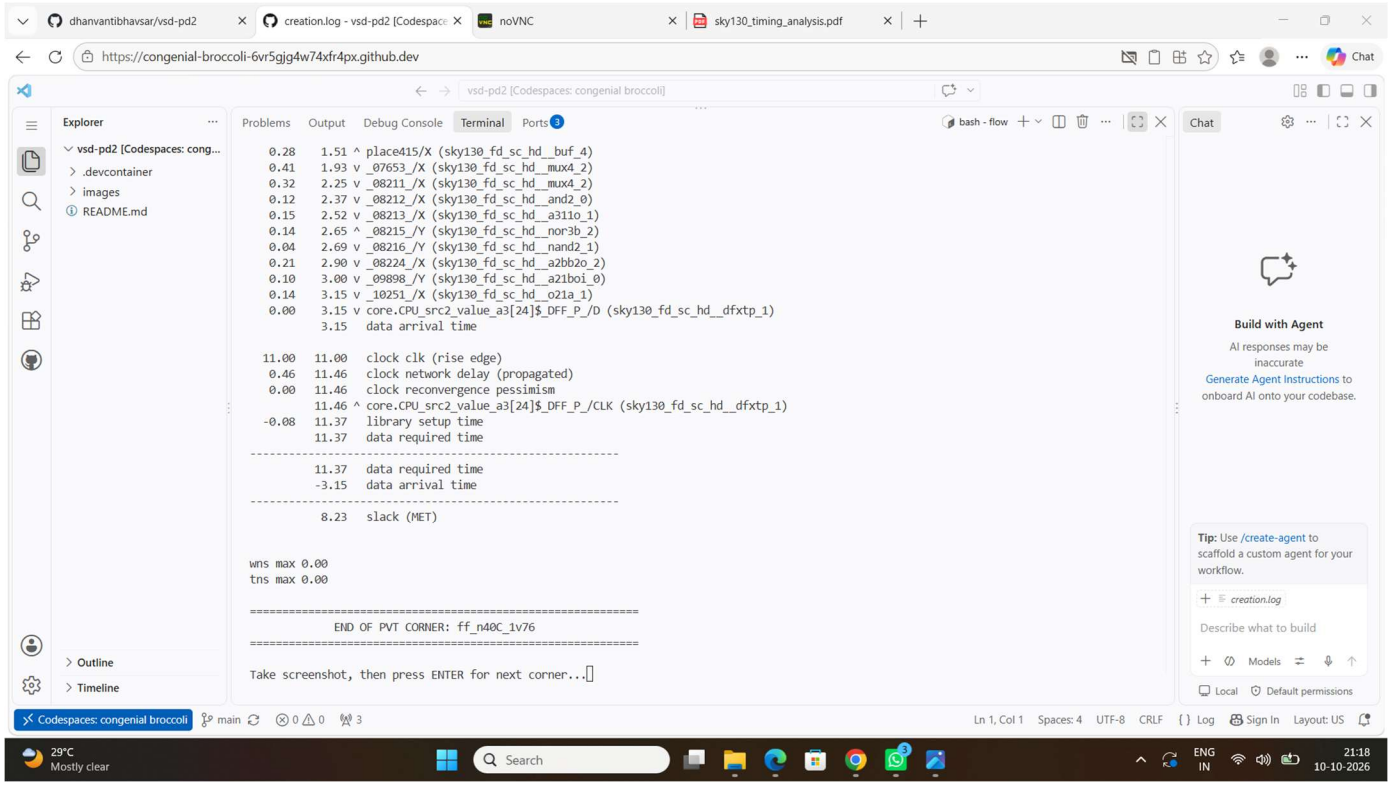
PVT Corner: ff_n40C_1v65

This represents the fast-fast (FF) process corner at −40°C and 1.65 V supply voltage.



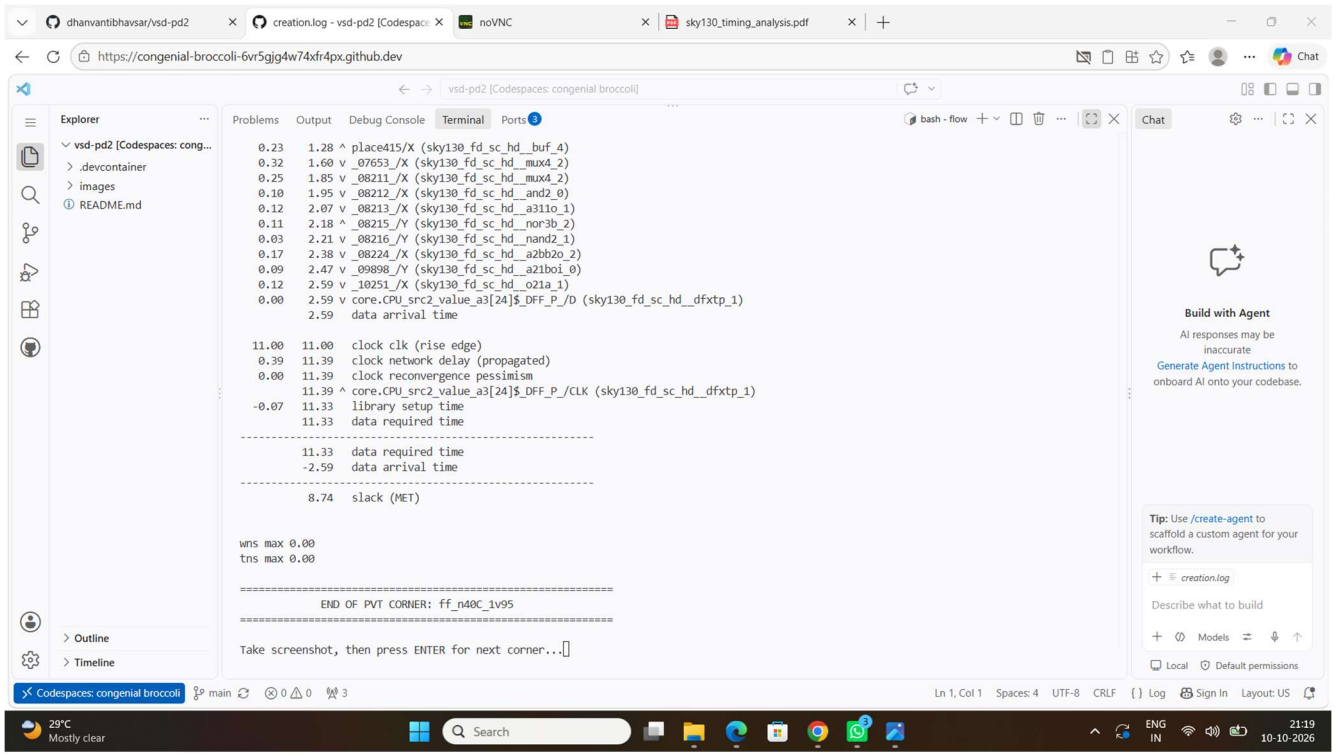
PVT Corner: ff_n40C_1v76

This represents the fast-fast (FF) process corner at −40°C and 1.76 V supply voltage.



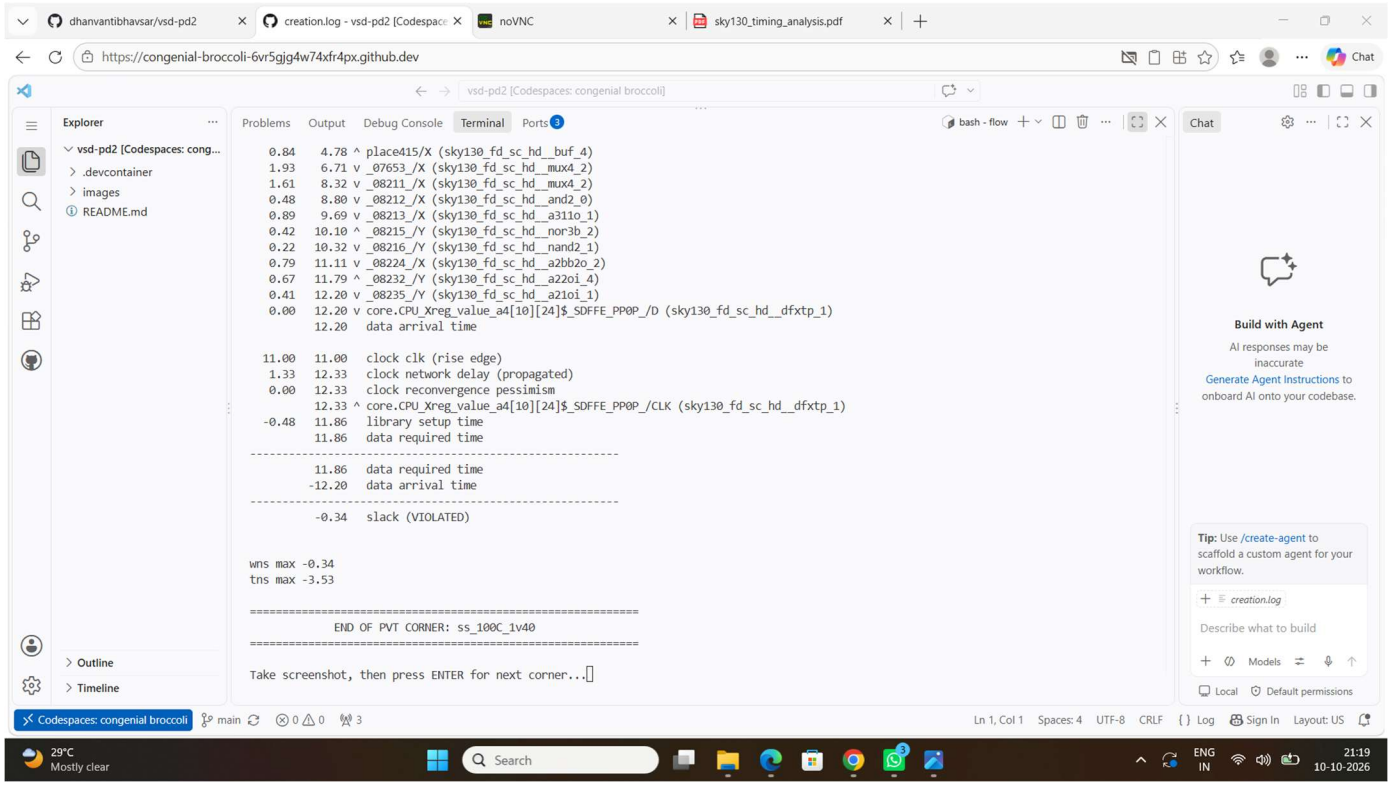
PVT Corner: ff_n40C_1v95

This represents the fast-fast (FF) process corner at −40°C and 1.95 V supply voltage.



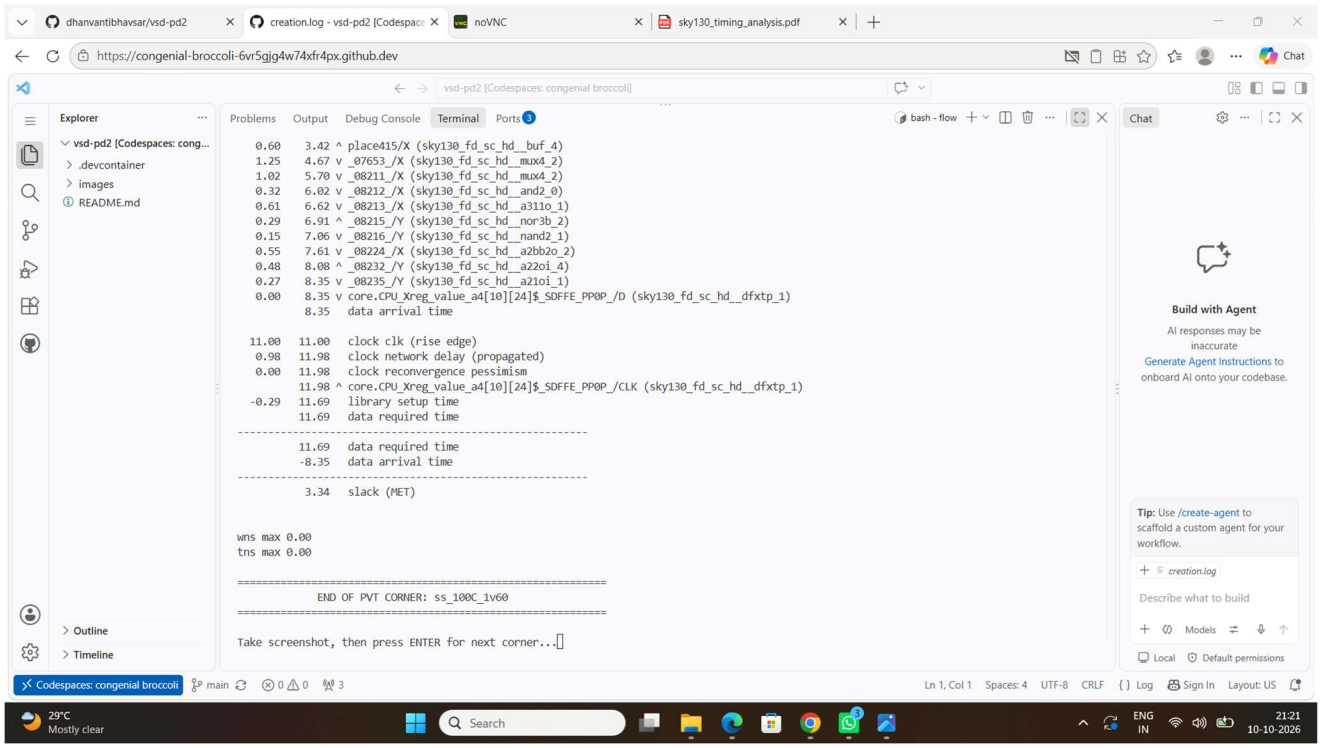
PVT Corner: ss_100C_1v40

This represents the slow-slow (SS) process corner at 100°C and 1.40 V supply voltage.



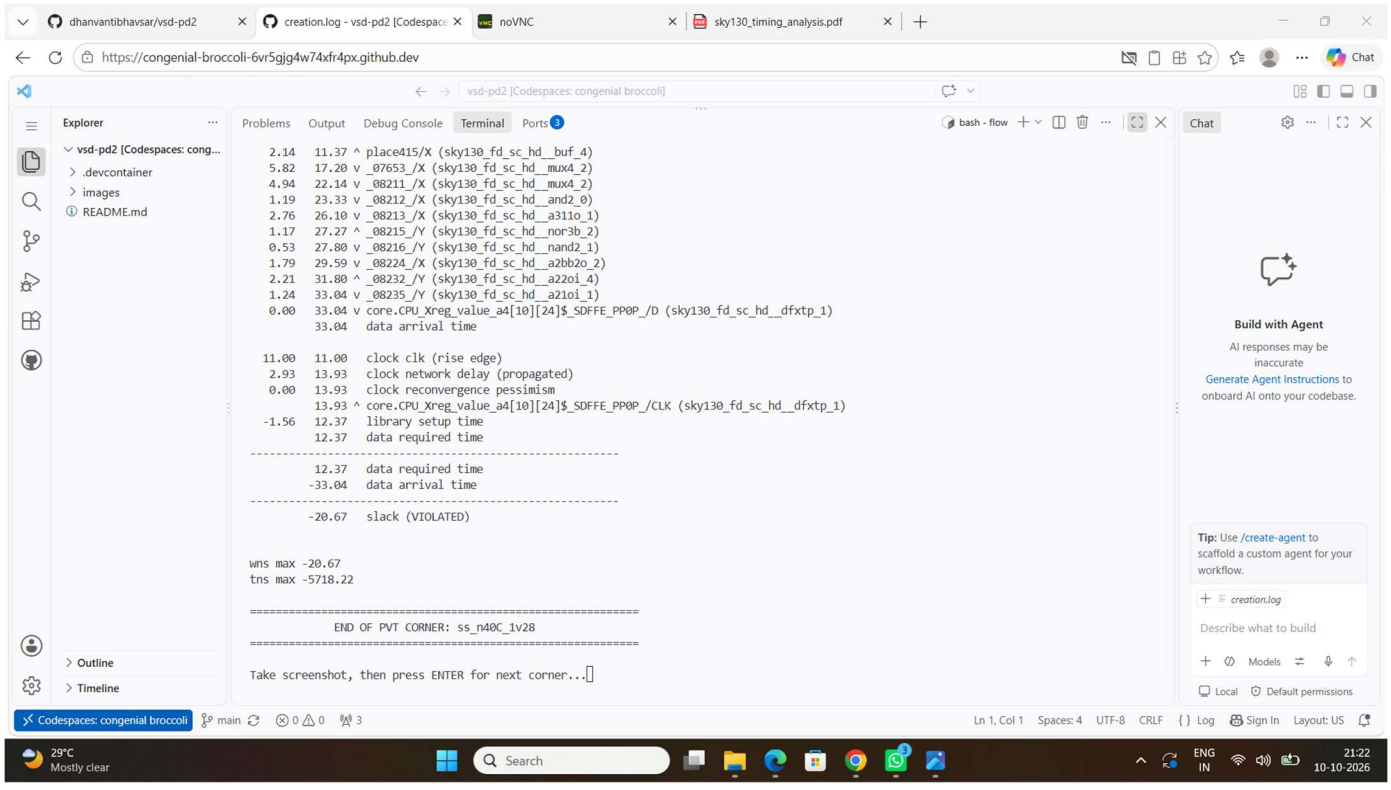
PVT Corner: ss_100C_1v60

This represents the slow-slow (SS) process corner at 100°C and 1.60 V supply voltage.



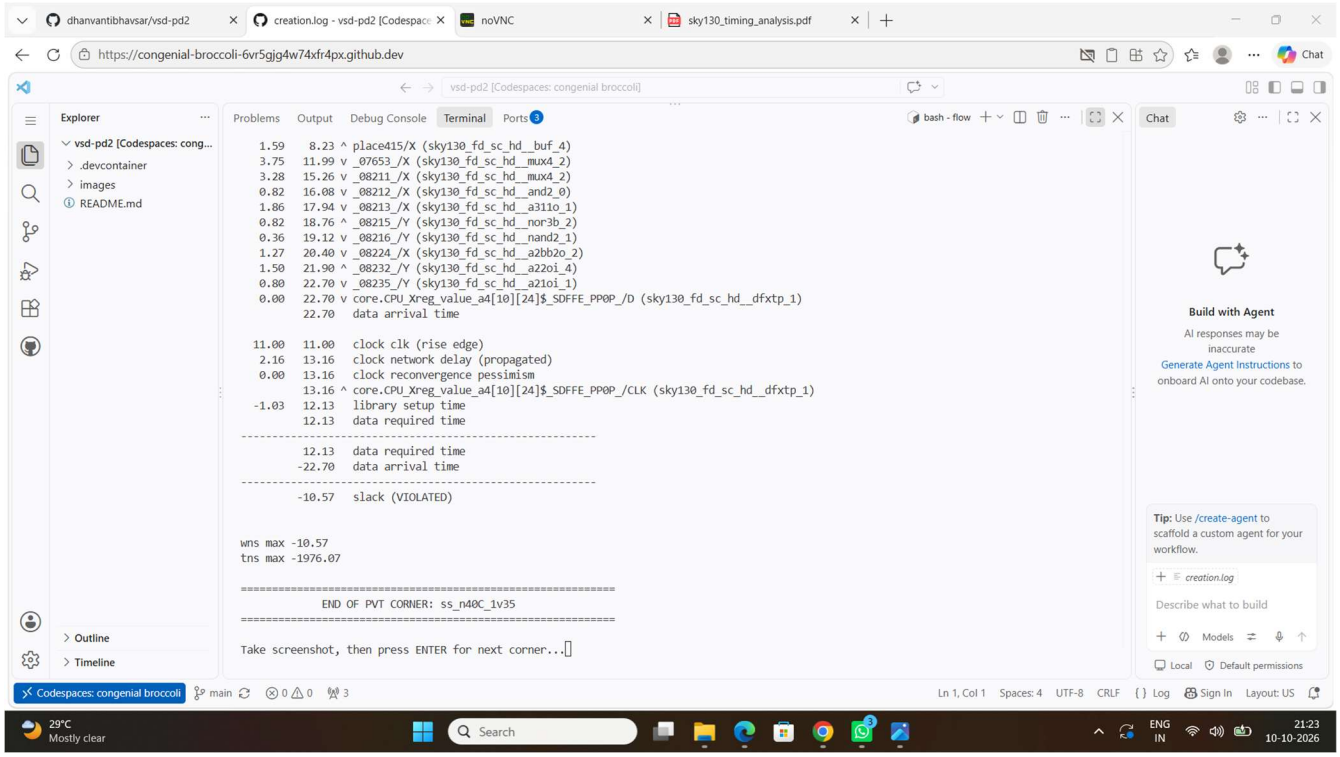
PVT Corner: ss_n40C_1v28

This represents the slow-slow (SS) process corner at −40°C and 1.28 V supply voltage.



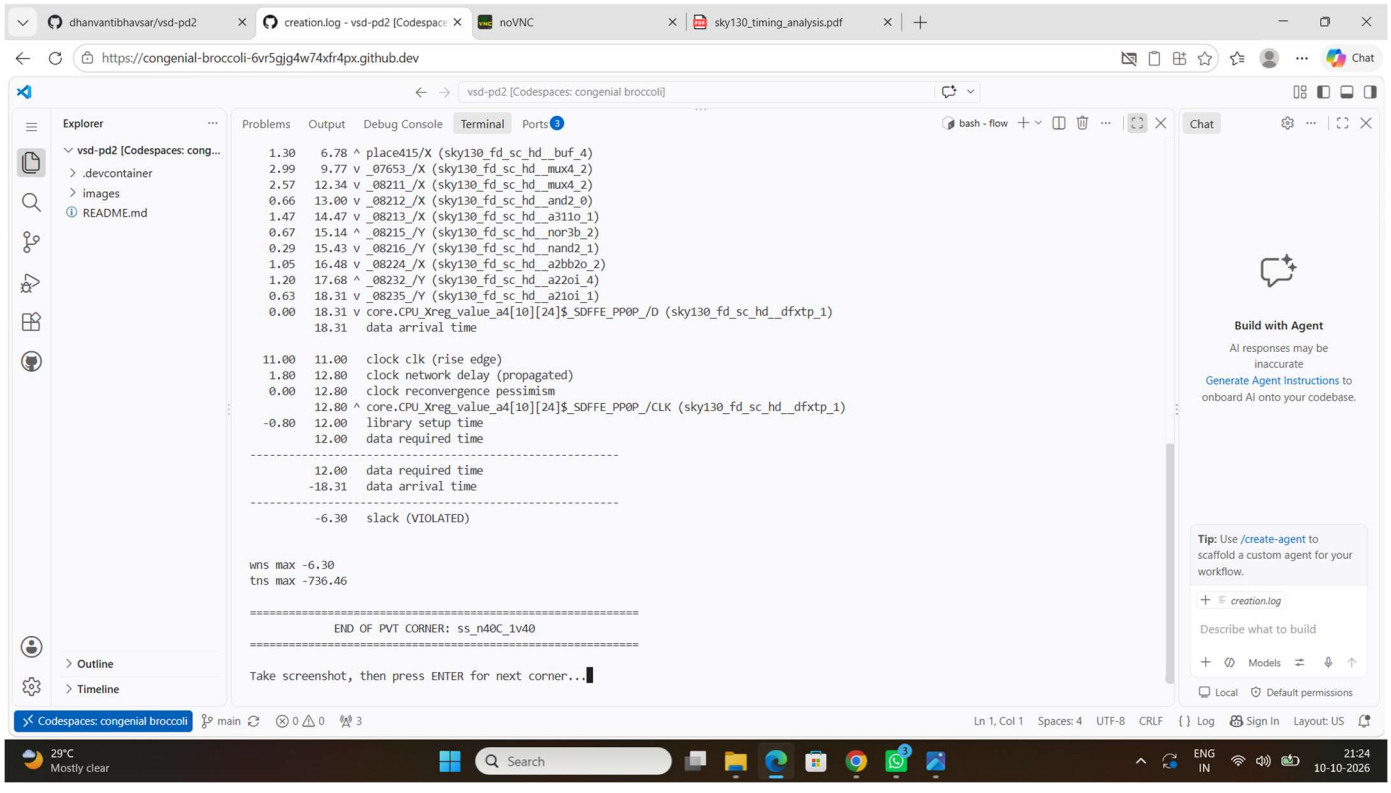
PVT Corner: ss_n40C_1v35

This represents the slow-slow (SS) process corner at −40°C and 1.35 V supply voltage.



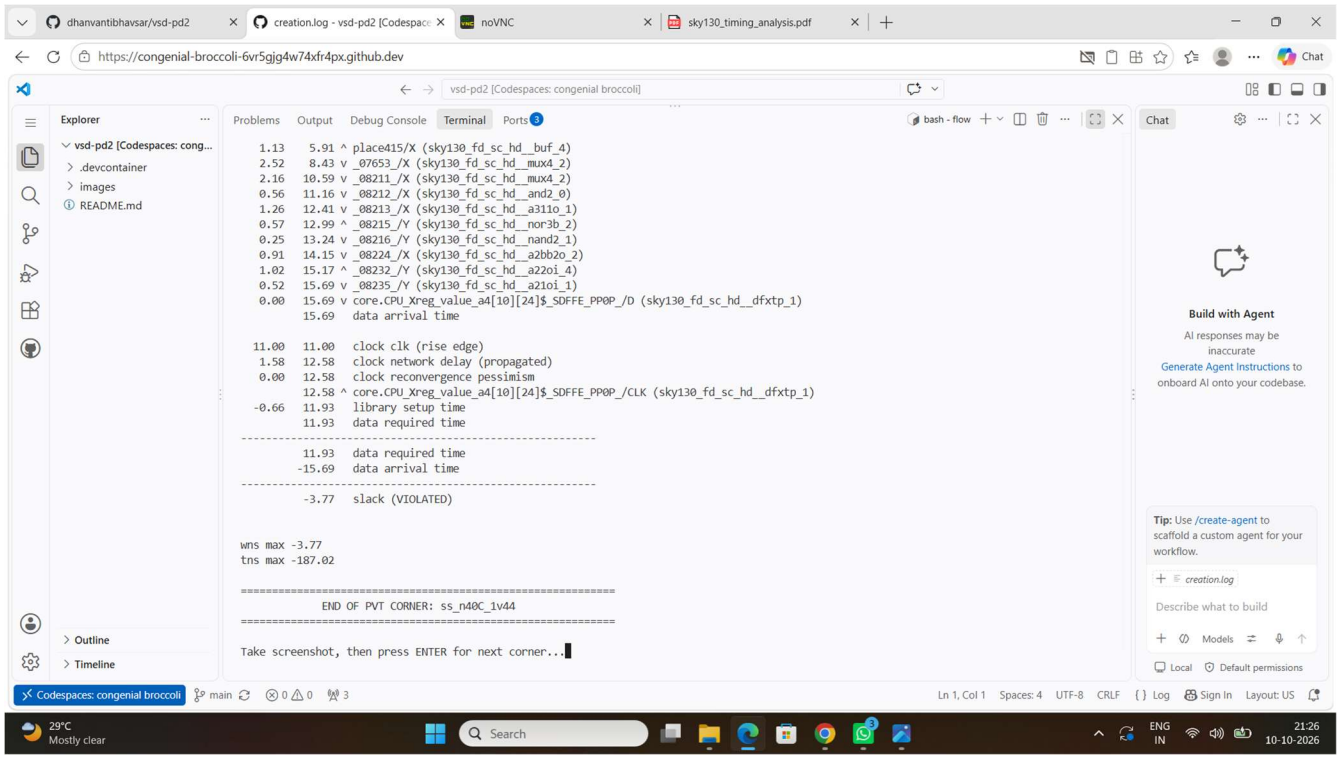
PVT Corner: ss_n40C_1v40

This represents the slow-slow (SS) process corner at −40°C and 1.40 V supply voltage.



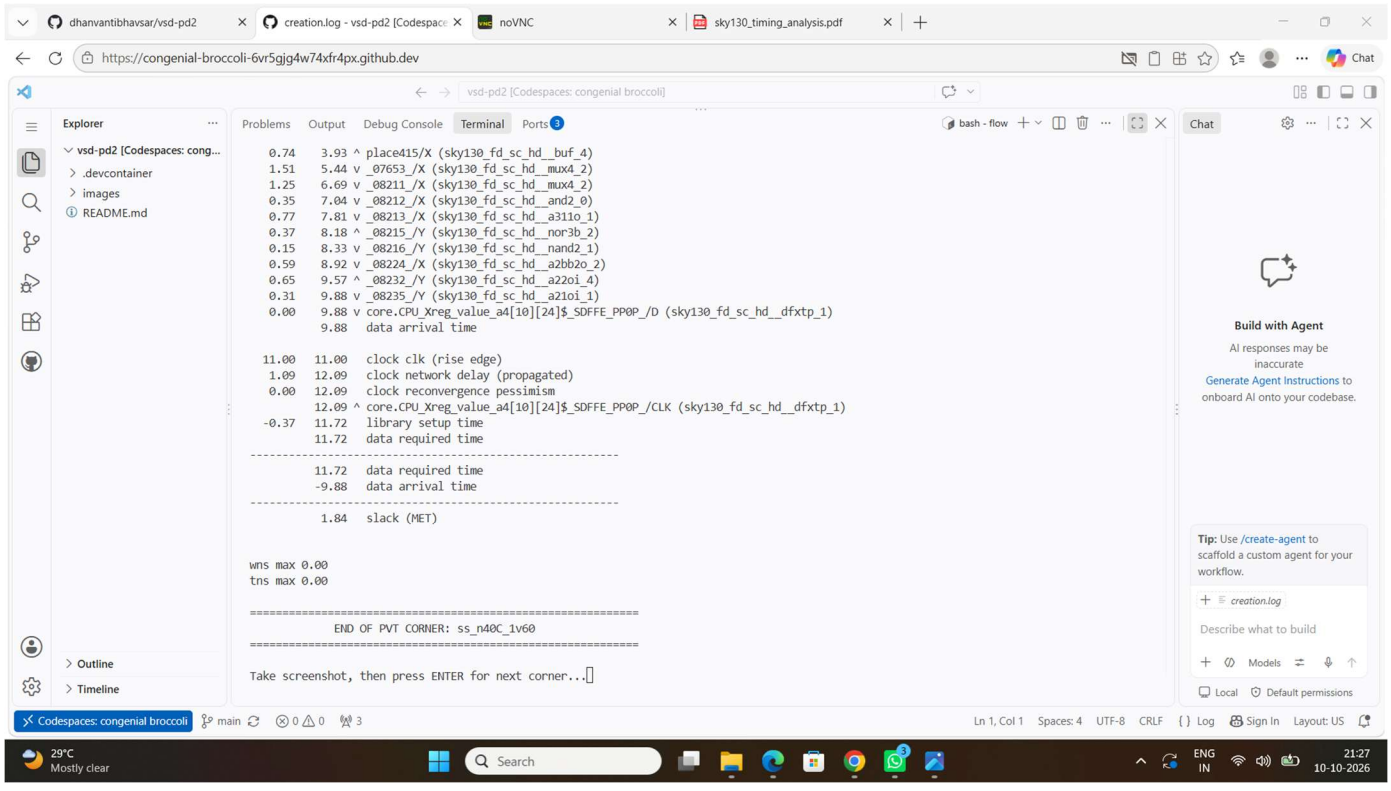
PVT Corner: ss_n40C_1v44

This represents the slow-slow (SS) process corner at −40°C and 1.44 V supply voltage.



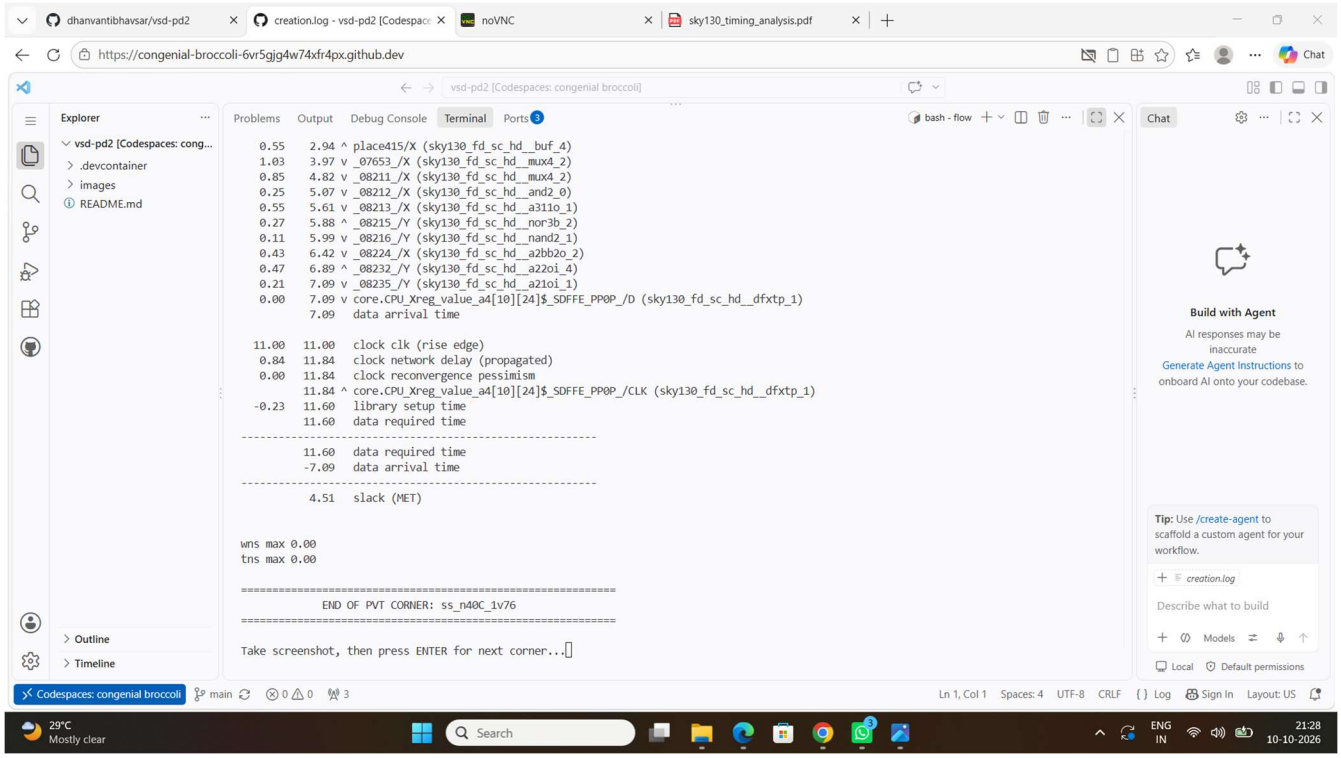
PVT Corner: ss_n40C_1v60

This represents the slow-slow (SS) process corner at −40°C and 1.60 V supply voltage.



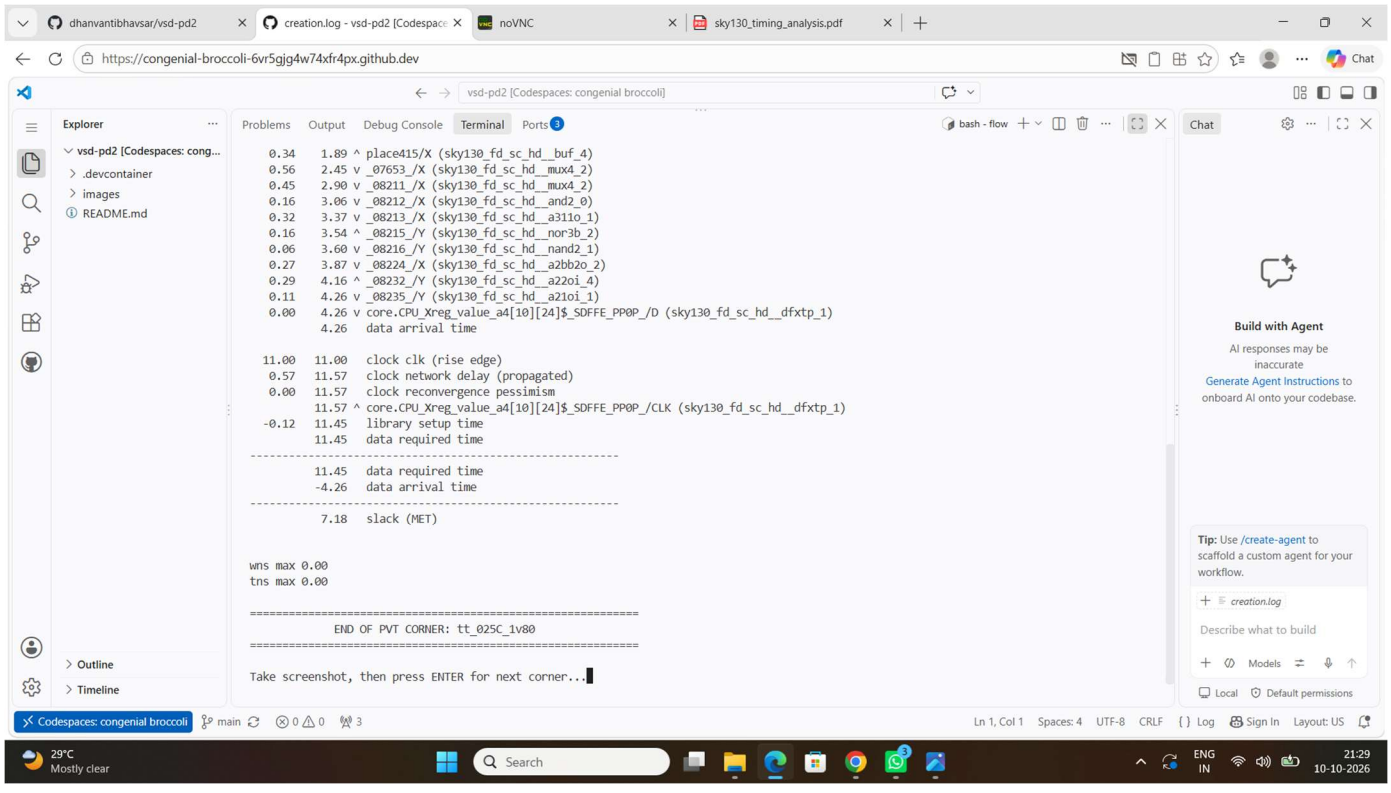
PVT Corner: ss_n40C_1v76

This represents the slow-slow (SS) process corner at −40°C and 1.76 V supply voltage.

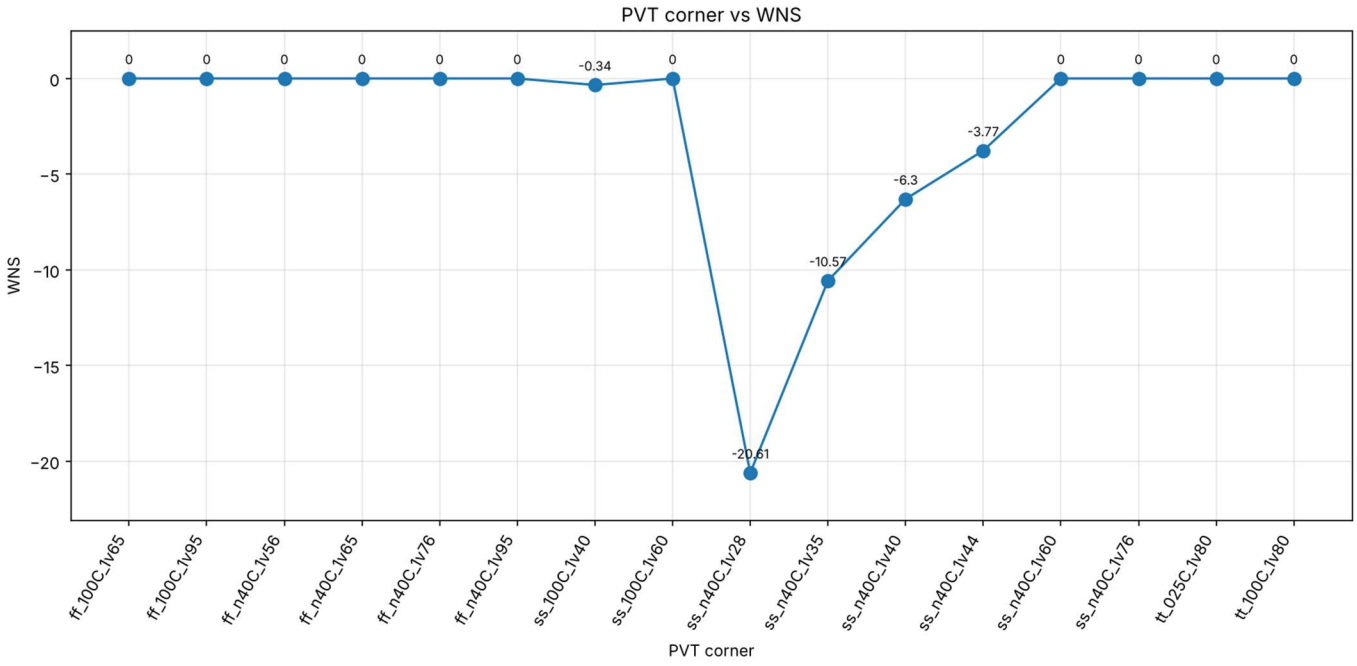


PVT Corner: tt_100C_1v80

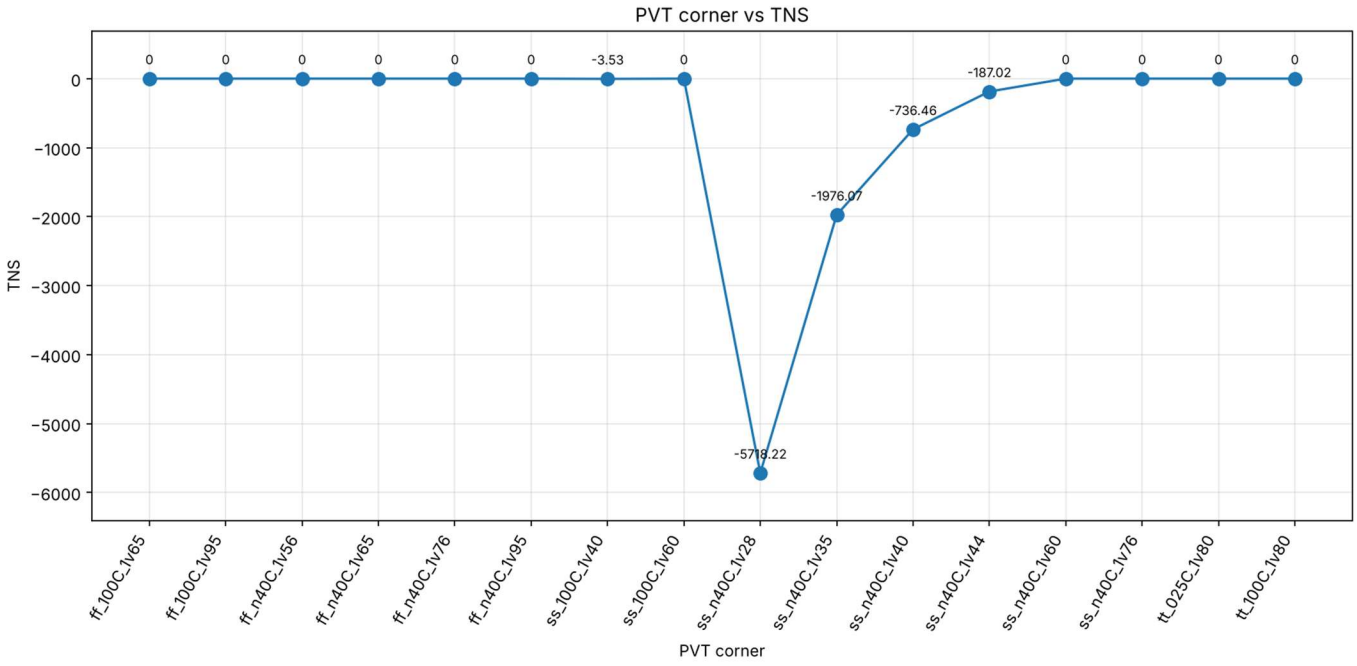
This represents the typical-typical (TT) process corner at 100°C and 1.80 V supply voltage.



PVT_Corner vs WNS Graph



PVT_Corner vs TNS Graph



5. Interpretation and Observations

The post-route WNS and TNS measurements are used to evaluate the timing performance of the routed VSDBabySoC design under different PVT conditions. Compared with pre-route analysis, post-route timing analysis can account for interconnect delays when extracted parasitic information is included.

The FF, TT, and SS corner results should be compared to identify how process conditions, voltage, and temperature affect timing. Negative WNS values indicate timing violations on at least one analyzed path, while negative TNS values indicate accumulated negative slack across multiple paths.

The most critical corner should be identified by comparing the actual post-route WNS and TNS results. The corner with the most negative WNS identifies the worst reported timing slack, while the most negative TNS indicates the largest accumulated negative slack.

The corresponding critical-path reports should be inspected to determine the contribution of cell delay, interconnect delay, clock behavior, and timing constraints. This information can guide subsequent timing optimization and help determine whether the design meets its timing requirements.

6. Conclusion

Post-Route Static Timing Analysis of the VSDBabySoC design was performed using OpenSTA across 16 SKY130 HD PVT corners. The analysis evaluates the timing behavior of the routed design under different process, voltage, and temperature conditions.

The individual timing reports and WNS/TNS comparison graphs provide a systematic way to identify timing-critical corners and investigate critical paths. The results can be used to guide timing optimization and evaluate timing closure across the required operating conditions.

The final assessment of timing performance must be based on the verified post-route WNS and TNS reports for all 16 PVT corners.